

Run: lr3e4-20ep

Log-log loss curves

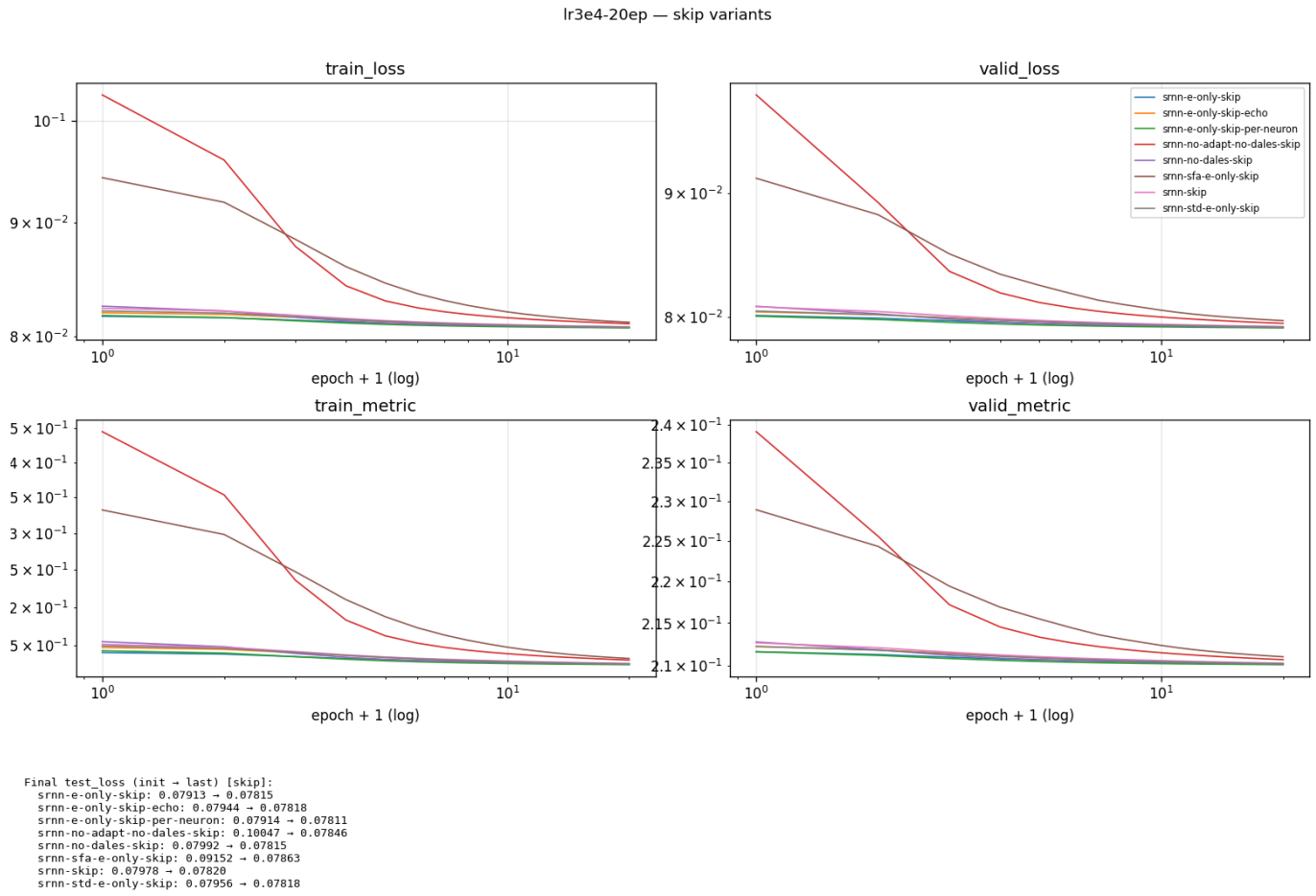


Figure 1: skip variants — log-log loss/metric

srnn-e-only-skip

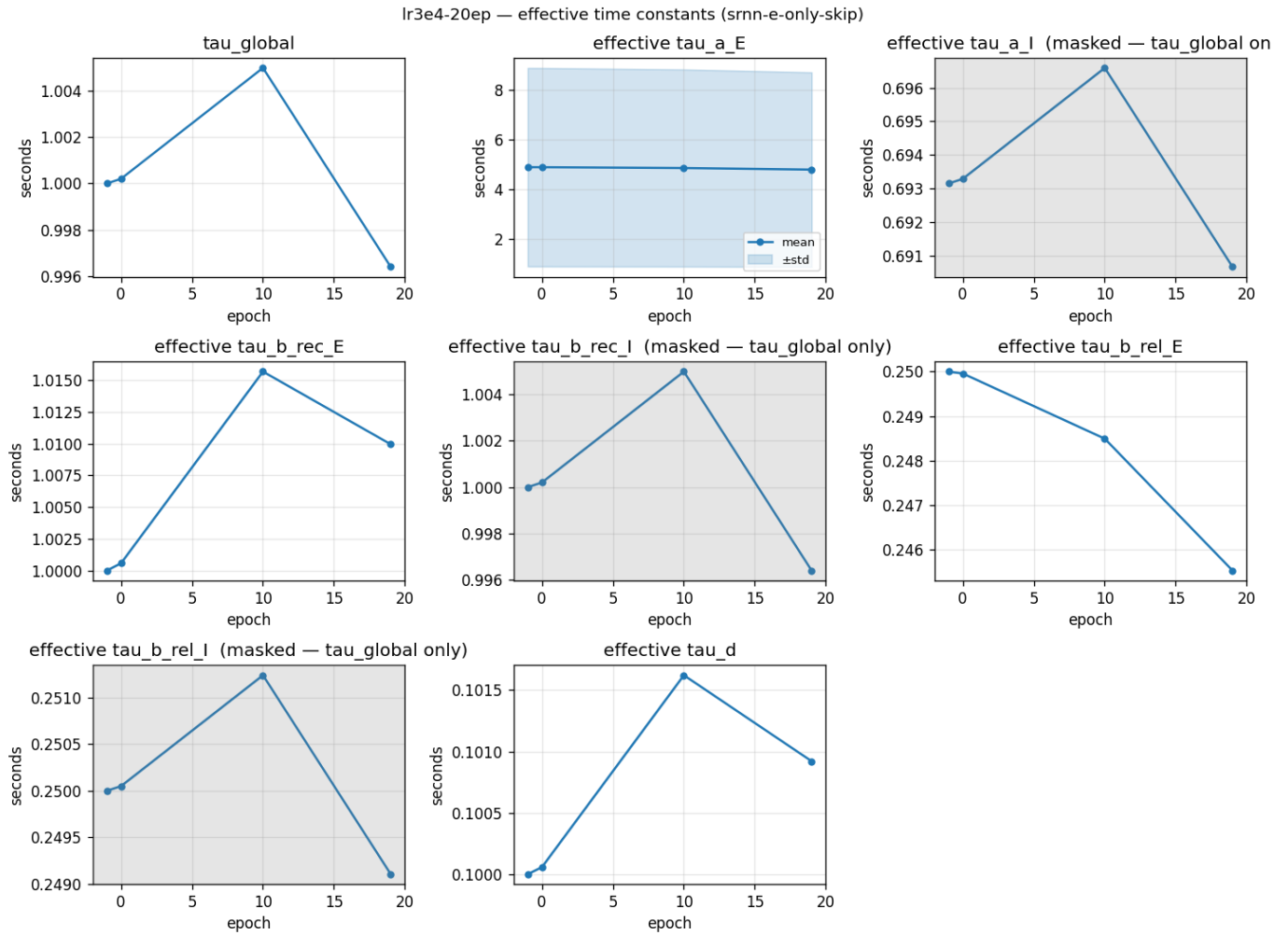


Figure 2: Tau evolution

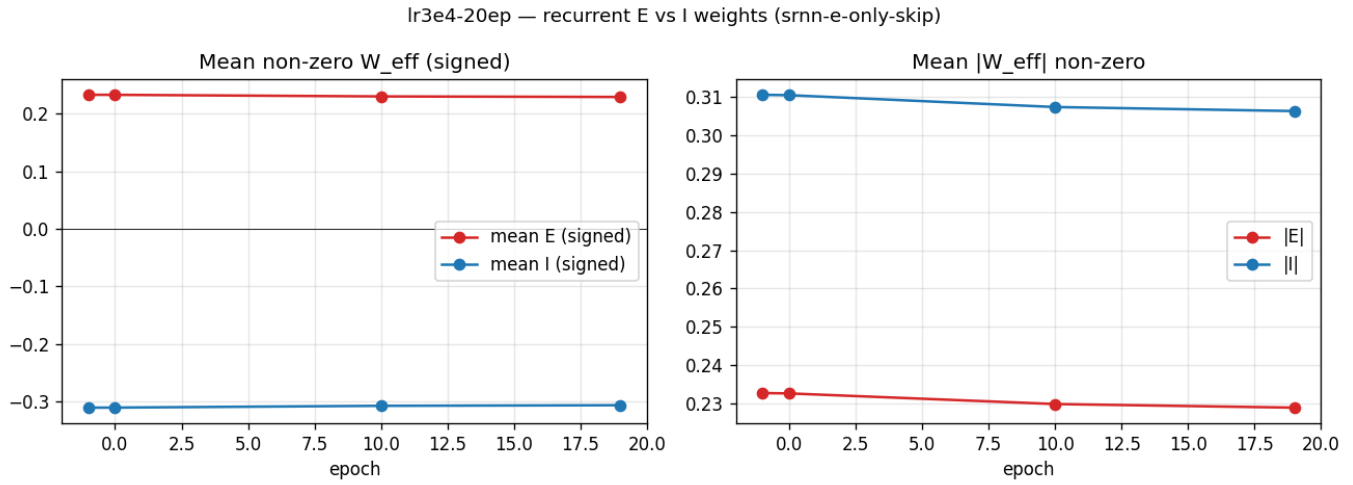


Figure 3: W_EI evolution

srnn-e-only-skip — parameter table

lr3e4-20ep - variant srnn-e-only-skip (k=0)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.99641	-	-	scalar
tau_d (s)	+0.1	+0.10092	0	0	300
tau_a_E (s)	+4.8777	+4.7793	3.996	3.915	450
tau_a_I (s)	+0.69315	+0.69066	0	0	450
tau_b_rec_E (s)	+1	+1.0099	0	0	150
tau_b_rel_E (s)	+0.25	+0.24552	0	0	150
tau_b_rec_I (s)	+1	+0.99641	0	0	150
tau_b_rel_I (s)	+0.25	+0.2491	0	0	150
W_eff (E src, signed)	+0.23267	+0.22885	0.07641	0.0752	15048
W_eff (I src, signed)	-0.31062	-0.30636	0.07724	0.0762	14977
W_eff (E src)	+0.23267	+0.22885	0.07641	0.0752	15048
W_eff (I src)	+0.31062	+0.30636	0.07724	0.0762	14977
W_in_eff (input neurons)	-0.0003386	-0.00033955	0.1012	0.09962	6675
c_E (SFA coupling)	+0.05	+0.050816	0	0	450
c_0_E (SFA offset)	+0	+0.017052	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36847	0	0	300
readout_weight	-0.00041422	+0.00041058	0.06568	0.05901	6675
readout_bias	+0	+0.0006861	0	0.003725	89
ic.ic (per-variant init)	+0.15932	+0.17082	0.4111	0.4078	1500

srnn-e-only-skip-per-neuron

lr3e4-20ep — effective time constants (srnn-e-only-skip-per-neuron)

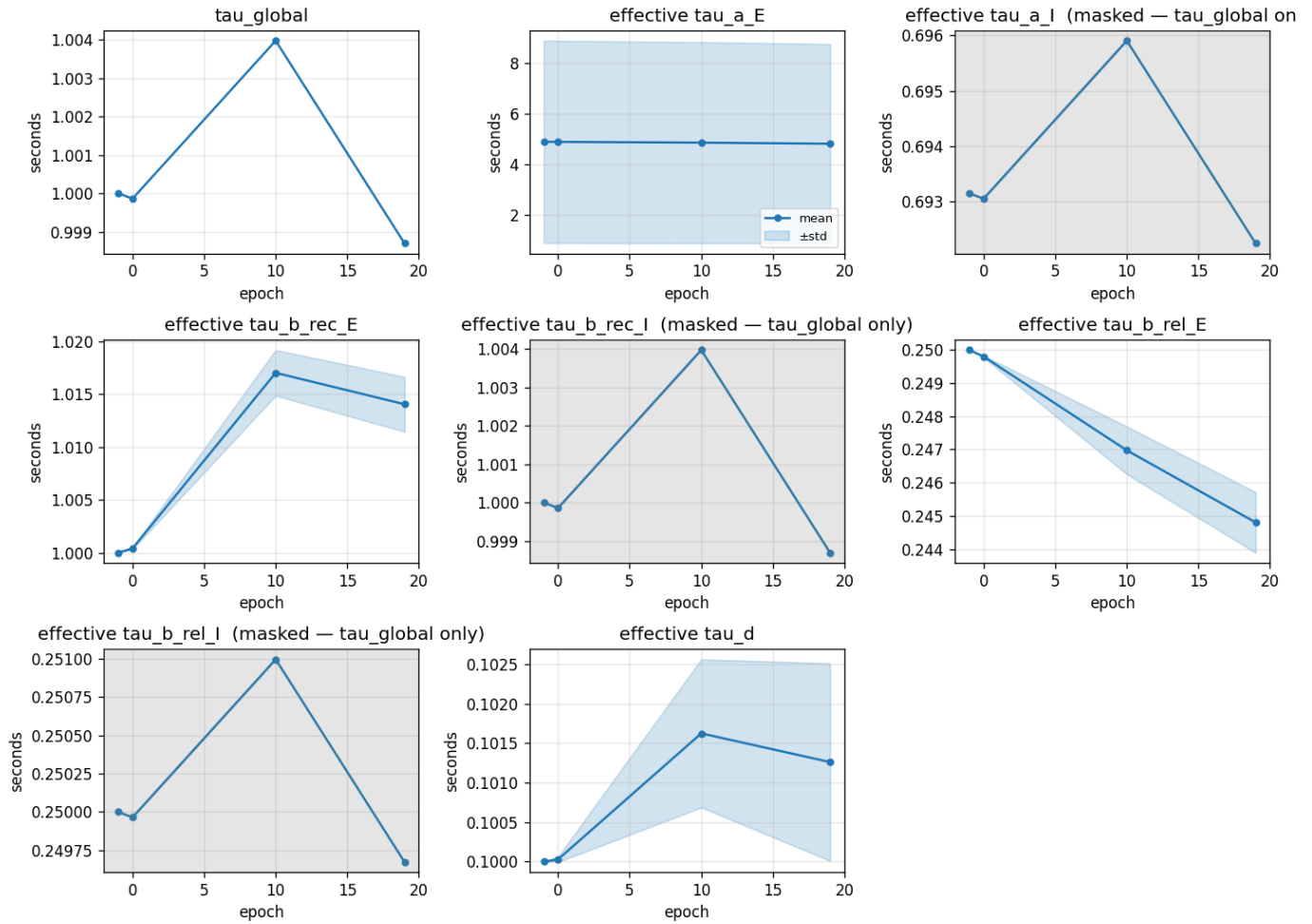


Figure 4: Tau evolution

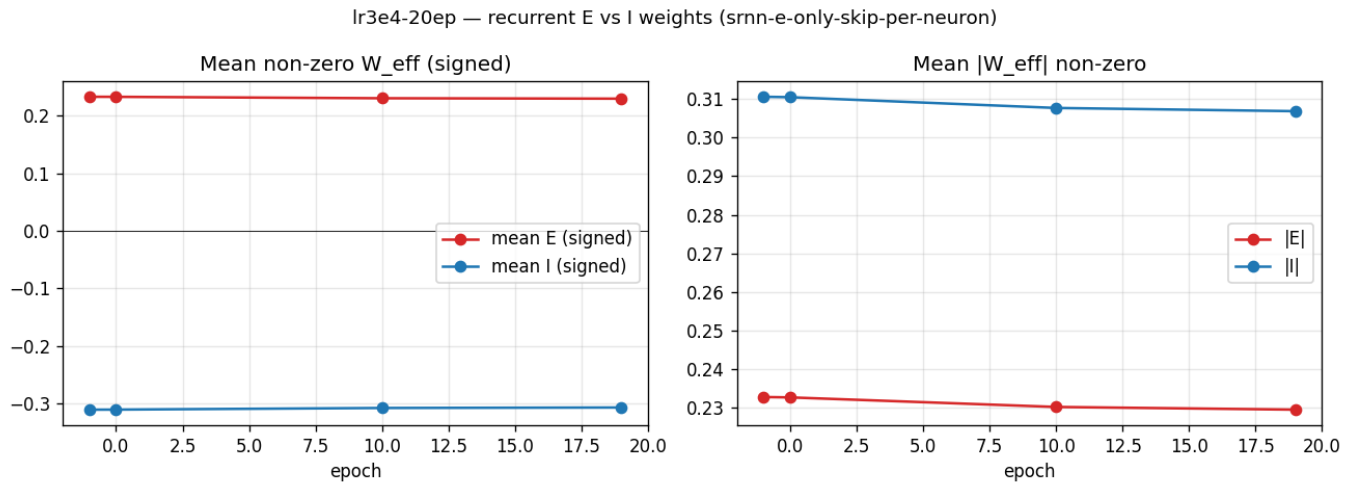


Figure 5: W_EI evolution

srnn-e-only-skip-per-neuron — parameter table

lr3e4-20ep - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.9987	-	-	scalar
tau_d (s)	+0.1	+0.10126	0	0.001252	300
tau_a_E (s)	+4.8777	+4.8026	3.996	3.936	450
tau_a_I (s)	+0.69315	+0.69224	0	0	450
tau_b_rec_E (s)	+1	+1.0141	0	0.002596	150
tau_b_rel_E (s)	+0.25	+0.2448	0	0.0009132	150
tau_b_rec_I (s)	+1	+0.9987	0	0	150
tau_b_rel_I (s)	+0.25	+0.24967	0	0	150
W_eff (E src, signed)	+0.23267	+0.22939	0.07641	0.07536	15048
W_eff (I src, signed)	-0.31062	-0.30688	0.07724	0.07631	14977
W_eff (E src)	+0.23267	+0.22939	0.07641	0.07536	15048
W_eff (I src)	+0.31062	+0.30688	0.07724	0.07631	14977
W_in_eff (input neurons)	-0.00084237	-0.0011027	0.1008	0.09777	6675
c_E (SFA coupling)	+0.05	+0.051195	0	0.0002591	450
c_0_E (SFA offset)	+0	+0.024719	0	0.004712	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.3674	0	0.01287	300
readout_weight	-0.0016154	-0.0015687	0.06624	0.06157	6675
readout_bias	+0	+0.00041623	0	0.00353	89
ic.ic (per-variant init)	+0.15938	+0.17378	0.4127	0.4152	1500

srnn-e-only-skip-echo

lr3e4-20ep — effective time constants (srnn-e-only-skip-echo)

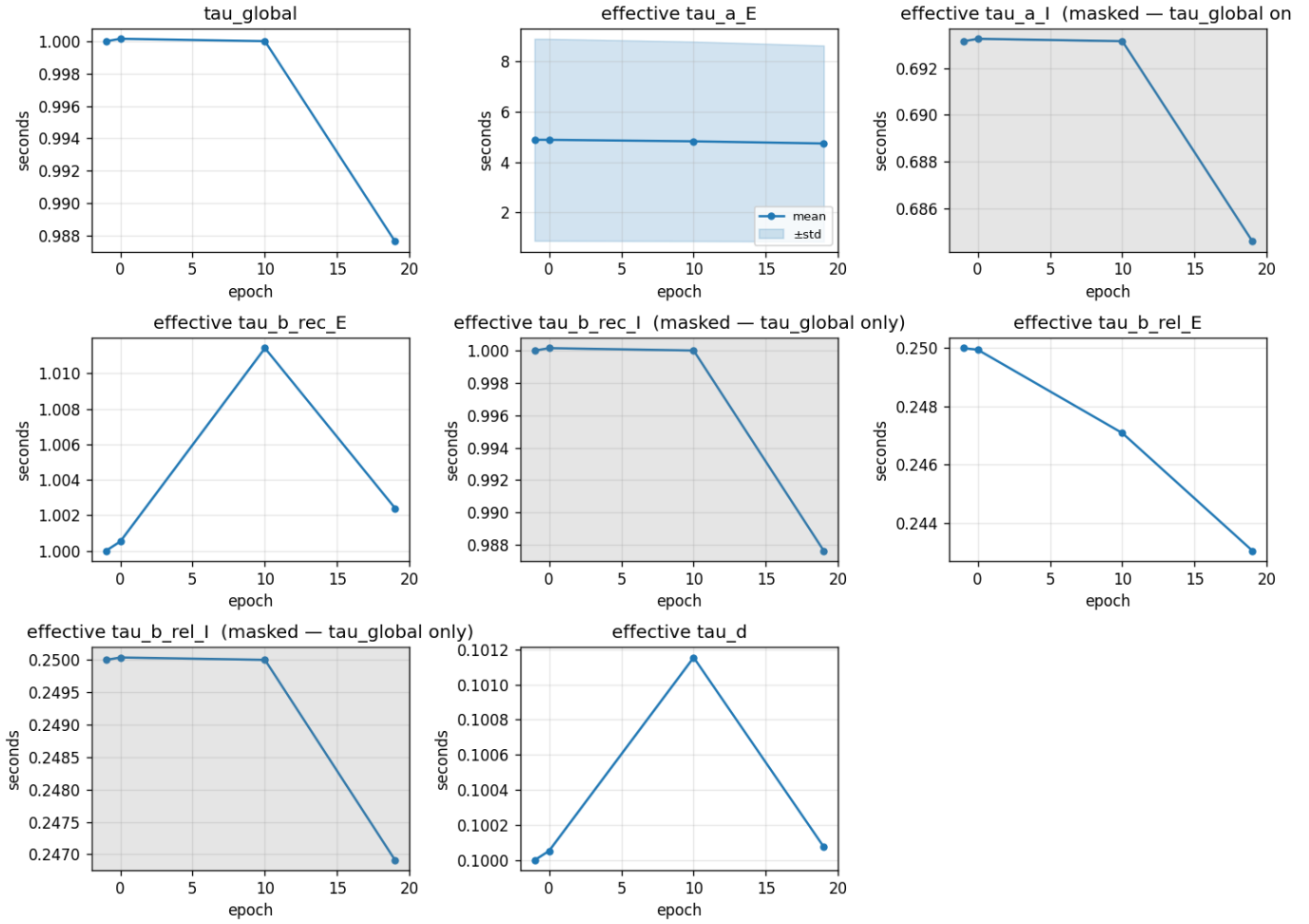


Figure 6: Tau evolution

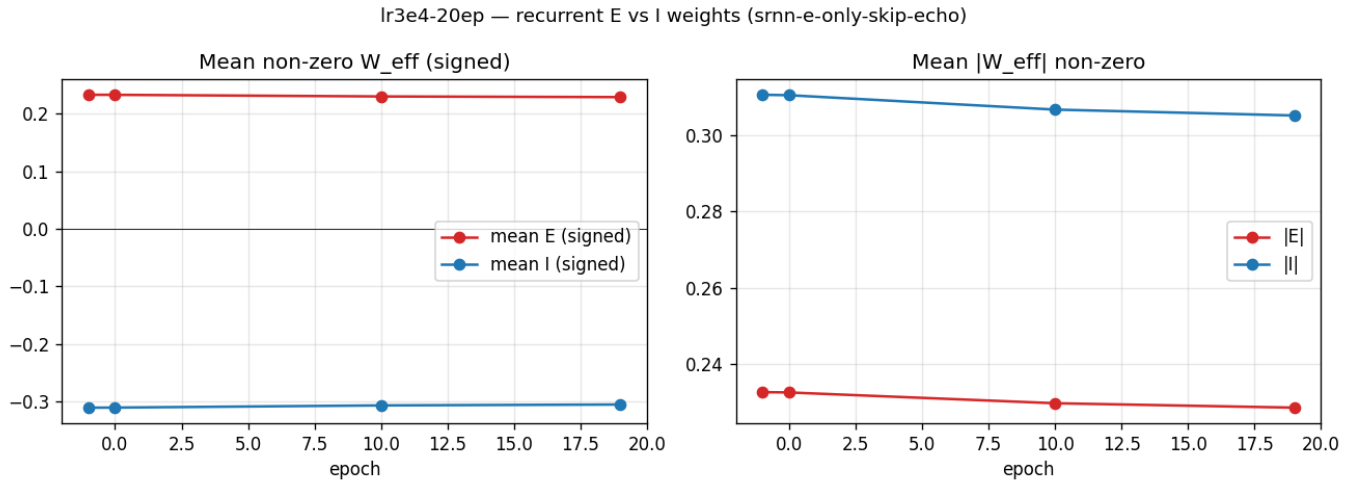


Figure 7: W_EI evolution

srnn-e-only-skip-echo — parameter table

lr3e4-20ep - variant srnn-e-only-skip-echo (k=2)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.98764	-	-	scalar
tau_d (s)	+0.1	+0.10008	0	0	300
tau_a_E (s)	+4.8777	+4.7312	3.996	3.876	450
tau_a_I (s)	+0.69315	+0.68458	0	0	450
tau_b_rec_E (s)	+1	+1.0024	0	0	150
tau_b_rel_E (s)	+0.25	+0.24305	0	0	150
tau_b_rec_I (s)	+1	+0.98764	0	0	150
tau_b_rel_I (s)	+0.25	+0.24691	0	0	150
W_eff (E src, signed)	+0.23267	+0.2286	0.07641	0.07508	15048
W_eff (I src, signed)	-0.31062	-0.30519	0.07724	0.07589	14977
W_eff (E src)	+0.23267	+0.2286	0.07641	0.07508	15048
W_eff (I src)	+0.31062	+0.30519	0.07724	0.07589	14977
W_in_eff (input neurons)	-0.00015382	-0.00019858	0.09934	0.09912	6675
c_E (SFA coupling)	+0.05	+0.050897	0	0	450
c_0_E (SFA offset)	+0	+0.018208	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36933	0	0	300
readout_weight	-0.00011373	-0.0011971	0.06687	0.06072	6675
readout_bias	+0	+0.00037265	0	0.005328	89
ic.ic (per-variant init)	+0.15962	+0.17043	0.4124	0.4065	1500

srnn-no-dales-skip

lr3e4-20ep — effective time constants (srnn-no-dales-skip)

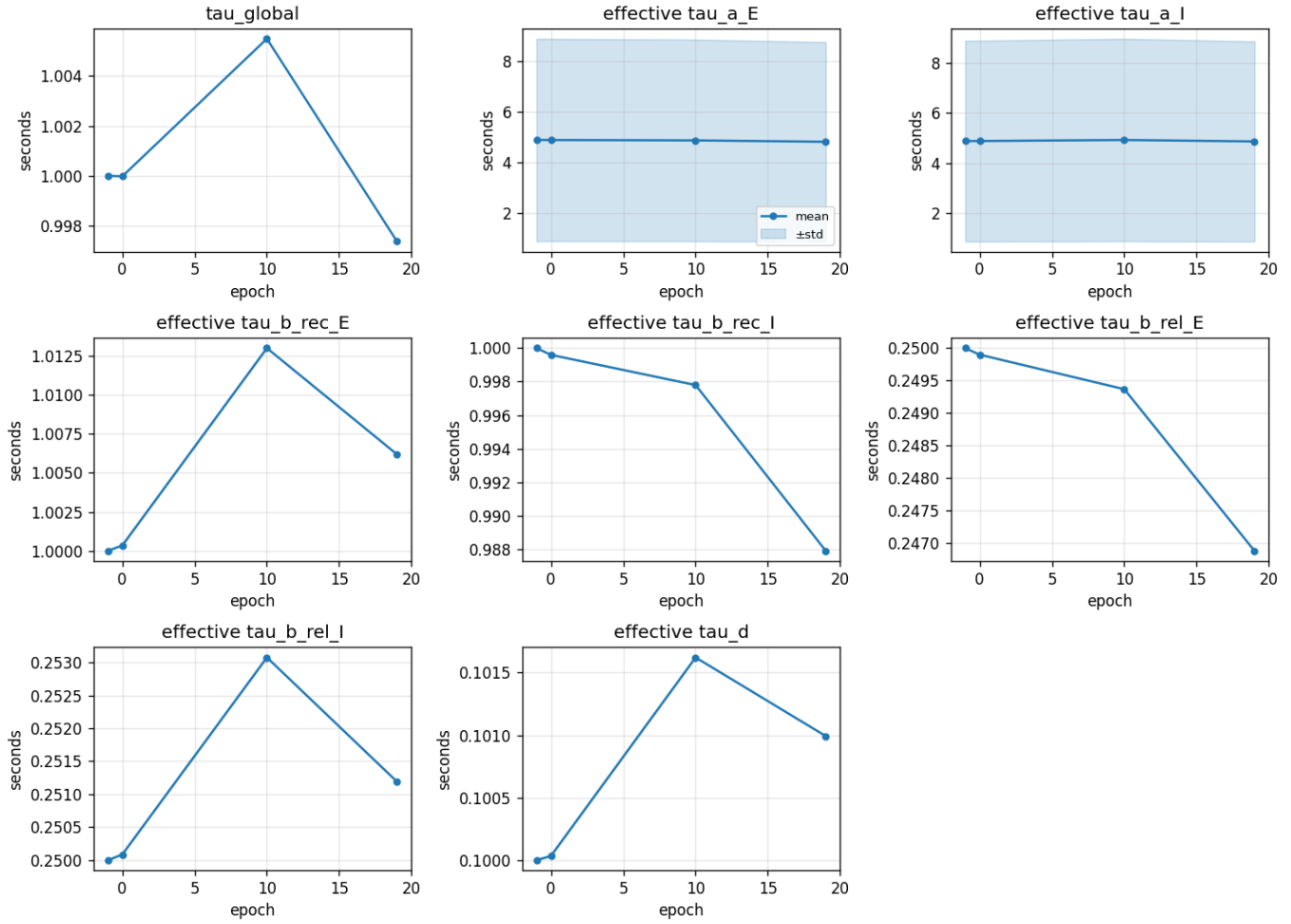
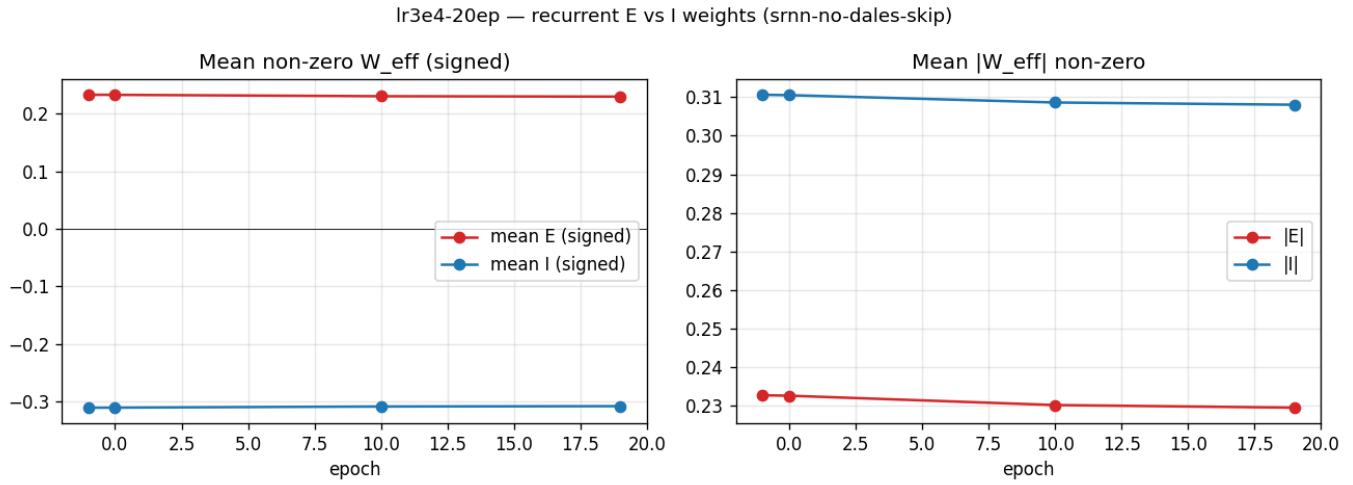


Figure 8: Tau evolution

Figure 9: W_{EI} evolution

srnn-no-dales-skip — parameter table

lr3e4-20ep - variant srnn-no-dales-skip (k=3)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.99736	-	-	scalar
tau_d (s)	+0.1	+0.10099	0	0	300
tau_a_E (s)	+4.8777	+4.8053	3.996	3.936	450
tau_a_I (s)	+4.8777	+4.8626	3.996	3.983	450
tau_b_rec_E (s)	+1	+1.0062	0	0	150
tau_b_rel_E (s)	+0.25	+0.24688	0	0	150
tau_b_rec_I (s)	+1	+0.9879	0	0	150
tau_b_rel_I (s)	+0.25	+0.25119	0	0	150
W_{eff} (E src, signed)	+0.2326	+0.22939	0.07663	0.07626	15048
W_{eff} (I src, signed)	-0.31062	-0.30803	0.07727	0.07678	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.22946	0.07641	0.07606	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30804	0.07724	0.07676	14977
$W_{\text{in_eff}}$ (input neurons)	-0.0011621	-0.00080286	0.09962	0.09653	6675
c_E (SFA coupling)	+0.05	+0.050473	0	0	450
c_0_E (SFA offset)	+0	+0.011531	0	0	450
c_I (SFA coupling)	+0.05	+0.049562	0	0	450
c_0_I (SFA offset)	+0	-0.010542	0	0	450
a_0 (threshold)	+0.35	+0.36292	0	0	300
readout_weight	-0.0016591	-0.0012728	0.06668	0.06284	6675
readout_bias	+0	+0.00088264	0	0.005516	89
ic.ic (per-variant init)	+0.16747	+0.16768	0.3627	0.3957	1500

srnn-no-adapt-no-dales-skip

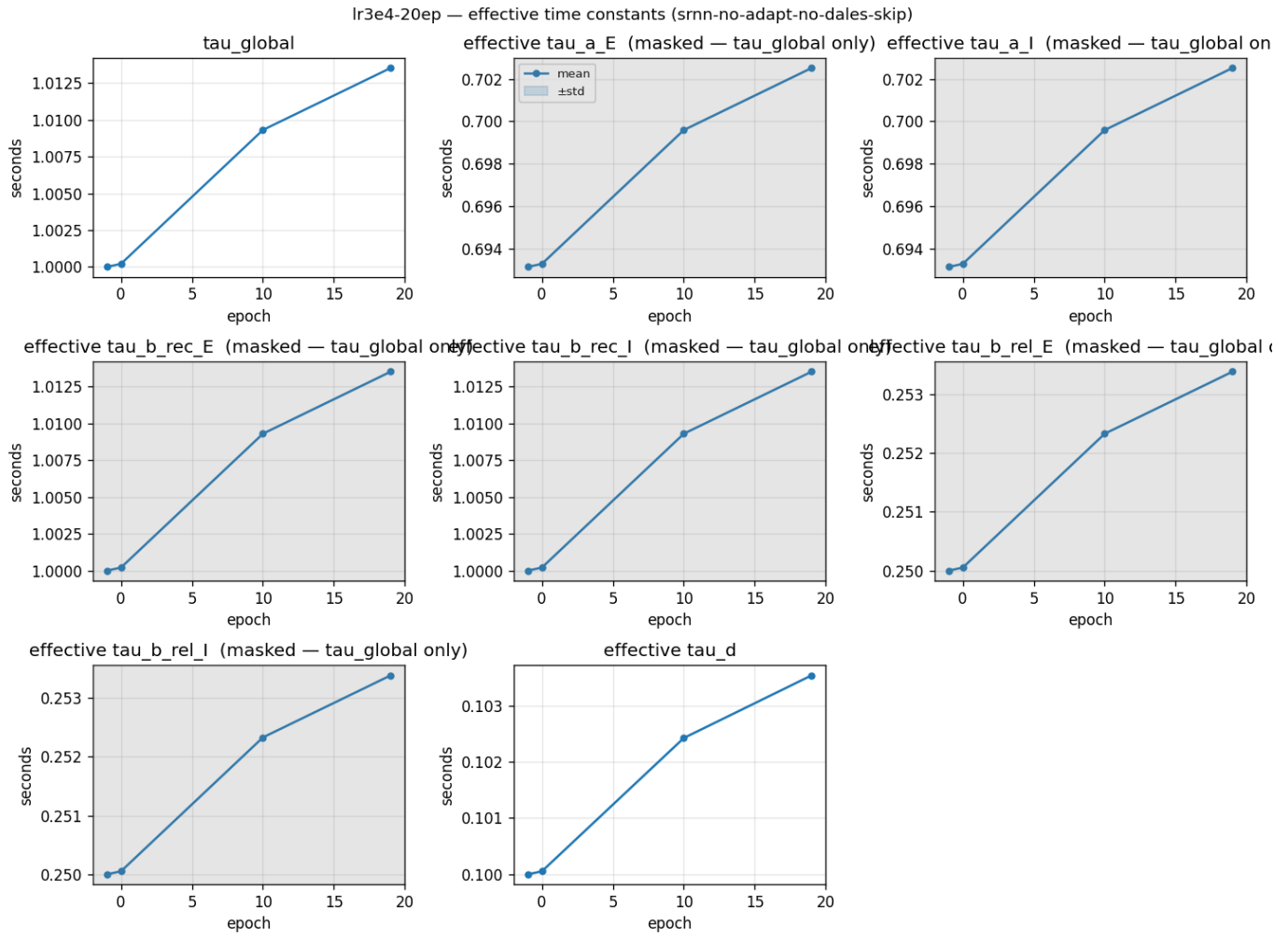
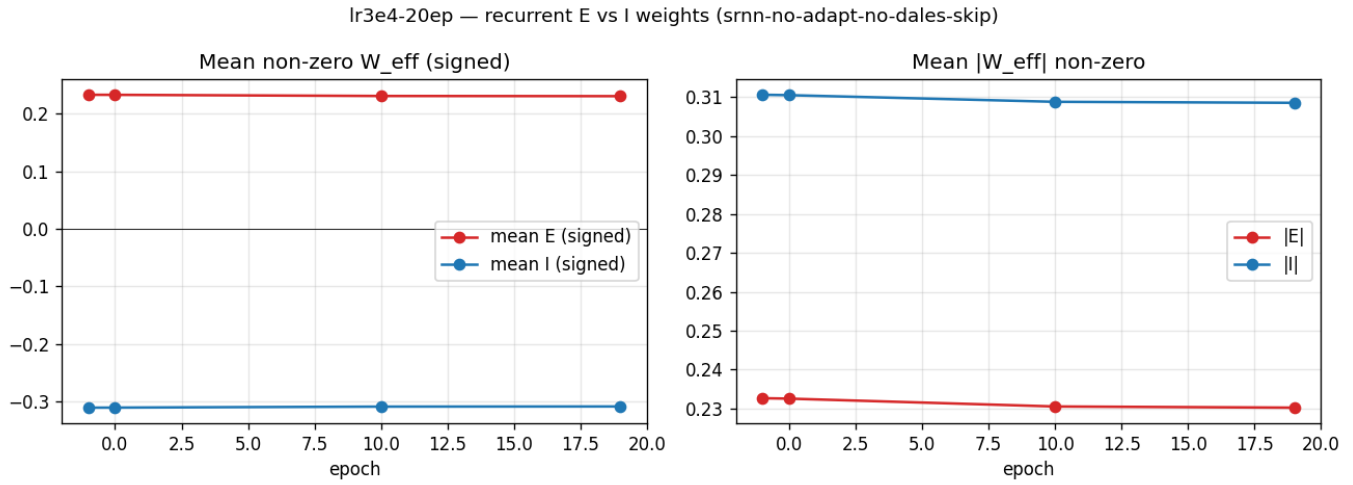


Figure 10: Tau evolution

Figure 11: W_{EI} evolution

srnn-no-adapt-no-dales-skip — parameter table

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# lr3e4-20ep - variant srnn-no-adapt-no-dales-skip (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
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effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+1.0135	-	-	scalar
tau_d (s)	+0.1	+0.10354	0	0	300
tau_a_E (s)	+0.69315	+0.70252	0	0	450
tau_a_I (s)	+0.69315	+0.70252	0	0	450
tau_b_rec_E (s)	+1	+1.0135	0	0	150
tau_b_rel_E (s)	+0.25	+0.25338	0	0	150
tau_b_rec_I (s)	+1	+1.0135	0	0	150
tau_b_rel_I (s)	+0.25	+0.25338	0	0	150
W_eff (E src, signed)	+0.2326	+0.23018	0.07663	0.07643	15048
W_eff (I src, signed)	-0.31062	-0.30855	0.07727	0.077	14977
W_eff (E src)	+0.23267	+0.23024	0.07641	0.07624	15048
W_eff (I src)	+0.31062	+0.30856	0.07724	0.07698	14977
W_in_eff (input neurons)	+0.00019944	+0.00032314	0.09994	0.09709	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_O_E (SFA offset)	+0	+0	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_O_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36142	0	0	300
readout_weight	+0.00079717	+0.0005836	0.06573	0.06295	6675
readout_bias	+0	-0.00017099	0	0.006122	89
ic.ic (per-variant init)	+0.14435	+0.15979	0.4566	0.4303	1500

srnn-skip

lr3e4-20ep — effective time constants (srnn-skip)

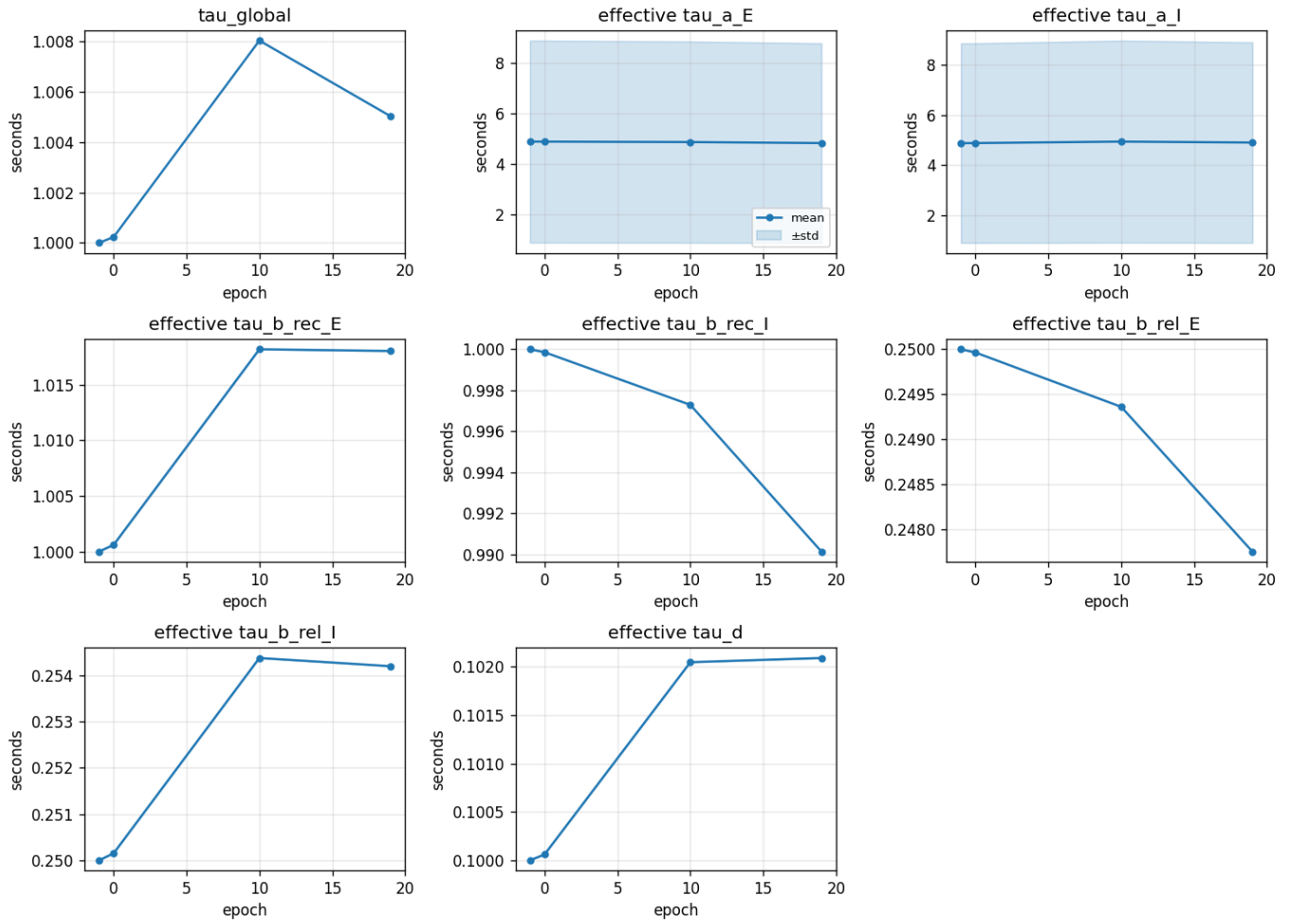


Figure 12: Tau evolution

lr3e4-20ep — recurrent E vs I weights (srnn-skip)

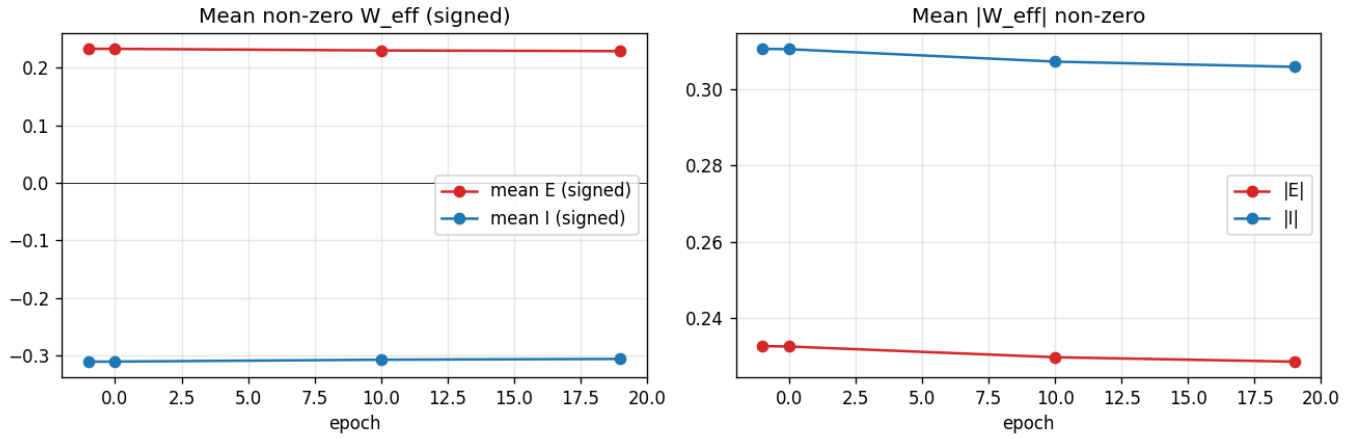


Figure 13: W_EI evolution

srnn-skip — parameter table

lr3e4-20ep - variant srnn-skip (k=5)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+1.005	-	-	scalar
tau_d (s)	+0.1	+0.10209	0	0	300
tau_a_E (s)	+4.8777	+4.8194	3.996	3.948	450
tau_a_I (s)	+4.8777	+4.9015	3.996	4.015	450
tau_b_rec_E (s)	+1	+1.018	0	0	150
tau_b_rel_E (s)	+0.25	+0.24775	0	0	150
tau_b_rec_I (s)	+1	+0.99014	0	0	150
tau_b_rel_I (s)	+0.25	+0.2542	0	0	150
W_eff (E src, signed)	+0.23267	+0.22861	0.07641	0.07512	15048
W_eff (I src, signed)	-0.31062	-0.30589	0.07724	0.07609	14977
W_eff (E src)	+0.23267	+0.22861	0.07641	0.07512	15048
W_eff (I src)	+0.31062	+0.30589	0.07724	0.07609	14977
W_in_eff (input neurons)	+0.0005819	+0.00038774	0.09954	0.09694	6675
c_E (SFA coupling)	+0.05	+0.050741	0	0	450
c_0_E (SFA offset)	+0	+0.016466	0	0	450
c_I (SFA coupling)	+0.05	+0.049347	0	0	450
c_0_I (SFA offset)	+0	-0.015484	0	0	450
a_0 (threshold)	+0.35	+0.36768	0	0	300
readout_weight	-0.00025143	-7.6565e-05	0.06686	0.06216	6675
readout_bias	+0	+0.0011882	0	0.007026	89
ic.ic (per-variant init)	+0.17068	+0.1695	0.3591	0.3823	1500

srnn-sfa-e-only-skip

lr3e4-20ep — effective time constants (srnn-sfa-e-only-skip)

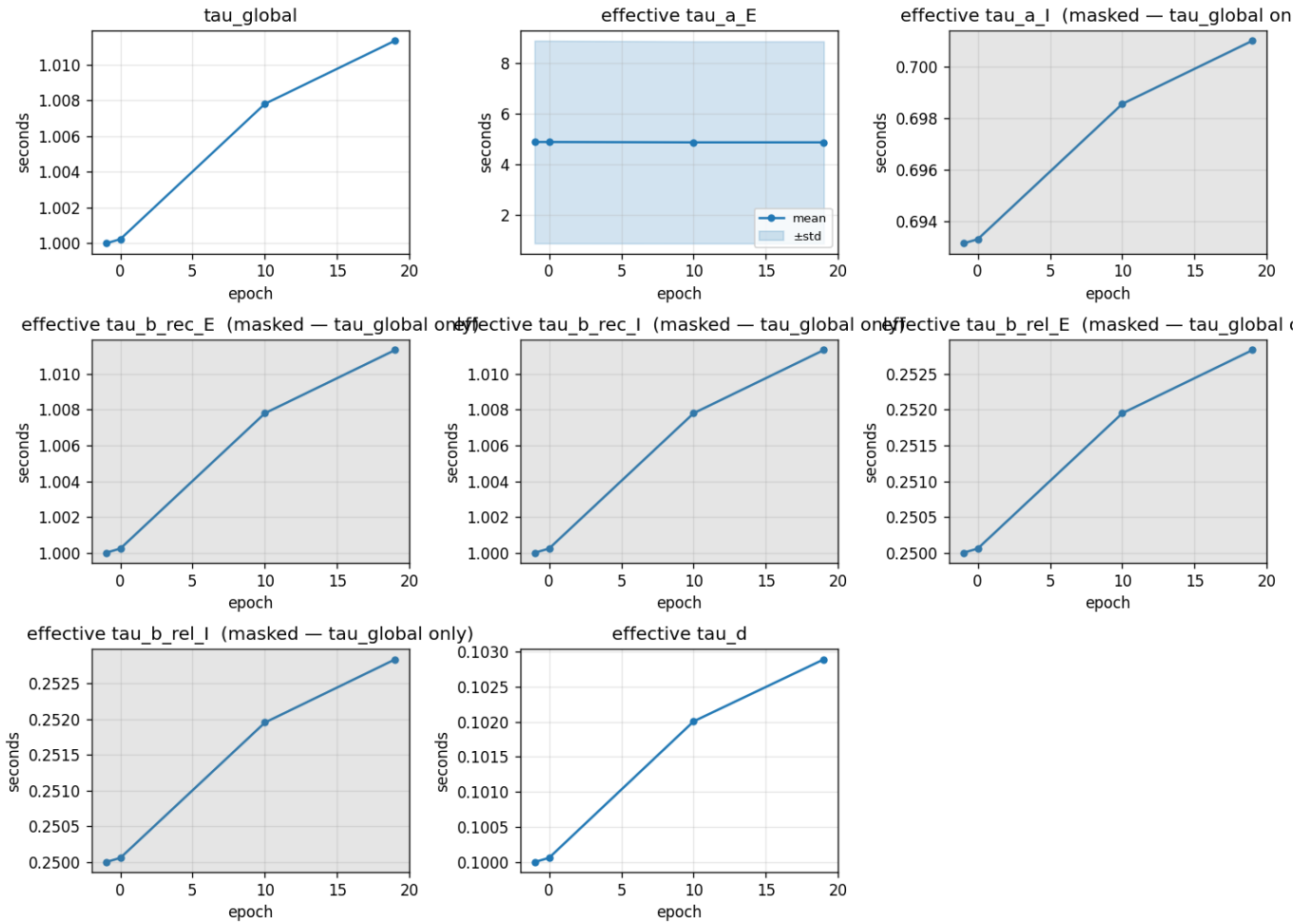
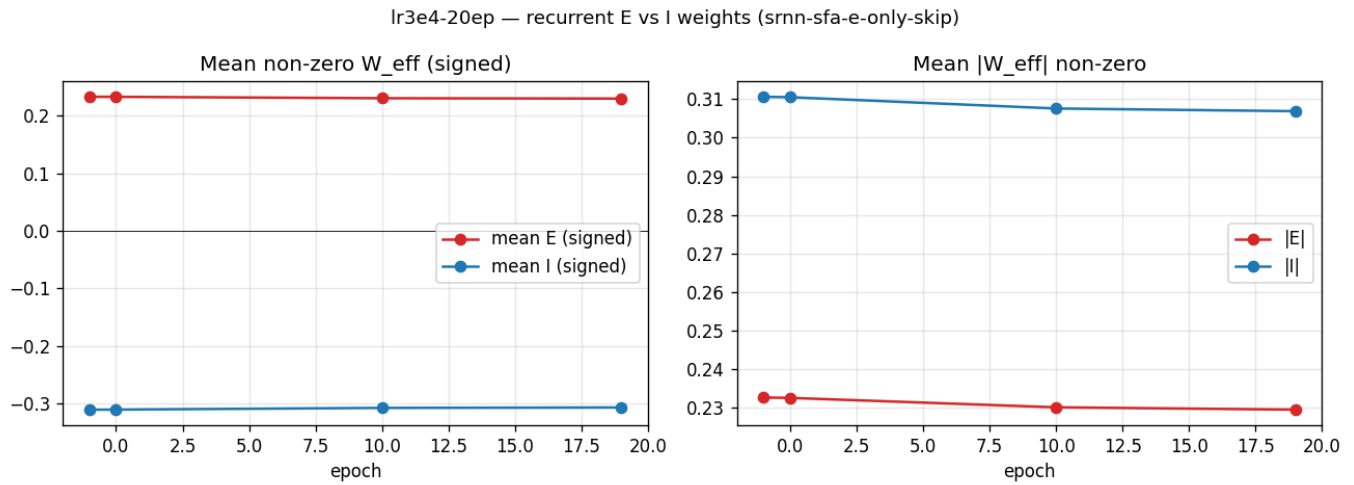


Figure 14: Tau evolution

Figure 15: W_{EI} evolution

srnn-sfa-e-only-skip — parameter table

lr3e4-20ep - variant srnn-sfa-e-only-skip (k=6)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+1.0113	-	-	scalar
tau_d (s)	+0.1	+0.10289	0	0	300
tau_a_E (s)	+4.8777	+4.8623	3.996	3.983	450
tau_a_I (s)	+0.69315	+0.701	0	0	450
tau_b_rec_E (s)	+1	+1.0113	0	0	150
tau_b_rel_E (s)	+0.25	+0.25283	0	0	150
tau_b_rec_I (s)	+1	+1.0113	0	0	150
tau_b_rel_I (s)	+0.25	+0.25283	0	0	150
W_{eff} (E src, signed)	+0.23267	+0.22953	0.07641	0.07543	15048
W_{eff} (I src, signed)	-0.31062	-0.30688	0.07724	0.07632	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.22953	0.07641	0.07543	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30688	0.07724	0.07632	14977
$W_{\text{in_eff}}$ (input neurons)	-0.00083129	-0.00065163	0.09974	0.09748	6675
c_E (SFA coupling)	+0.05	+0.050589	0	0	450
c_0_E (SFA offset)	+0	+0.014862	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36598	0	0	300
readout_weight	+0.001119	+0.00010576	0.06703	0.06326	6675
readout_bias	+0	-0.0010375	0	0.007758	89
ic.ic (per-variant init)	+0.16079	+0.17166	0.4384	0.4265	1500

srnn-std-e-only-skip

lr3e4-20ep — effective time constants (srnn-std-e-only-skip)

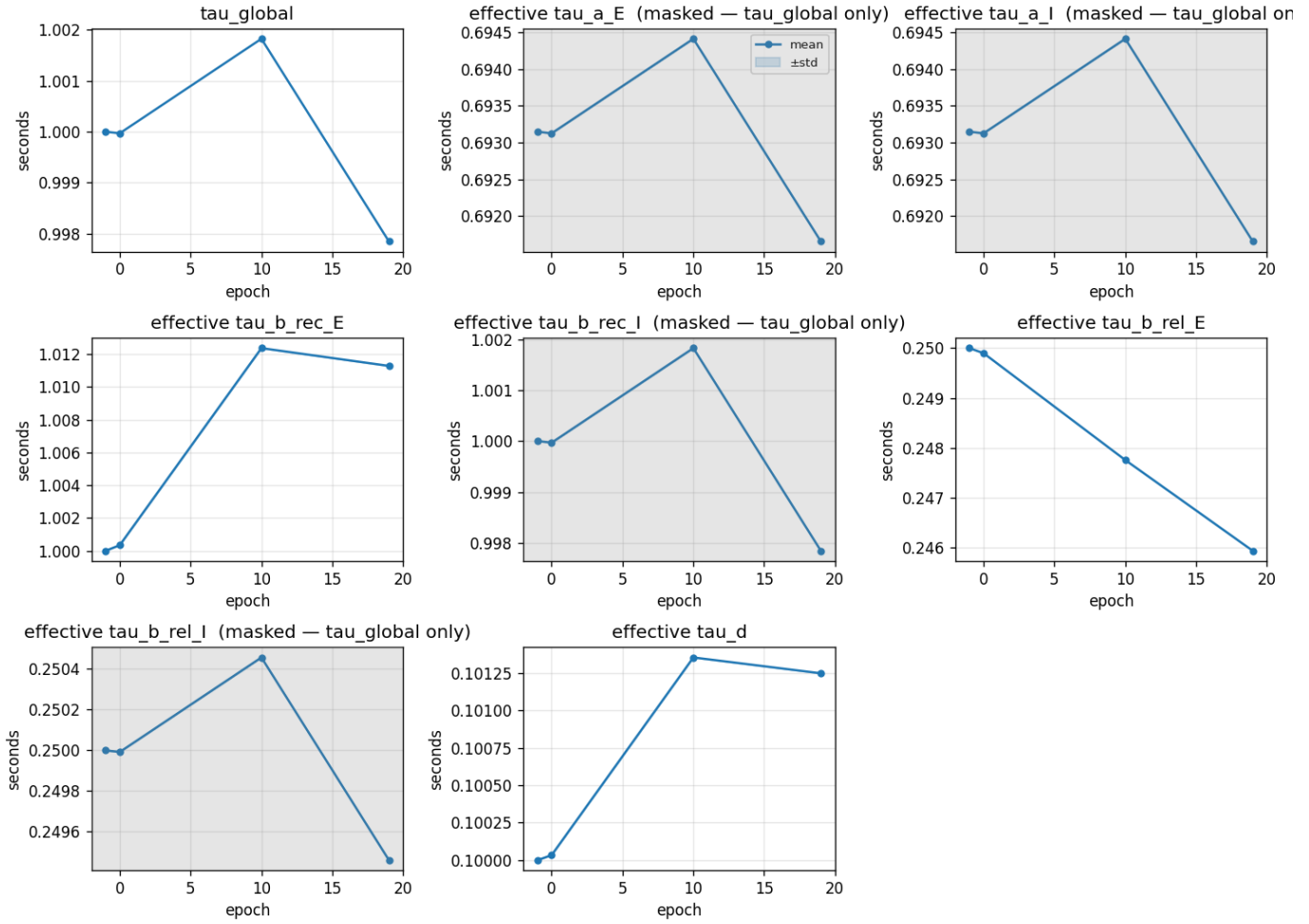
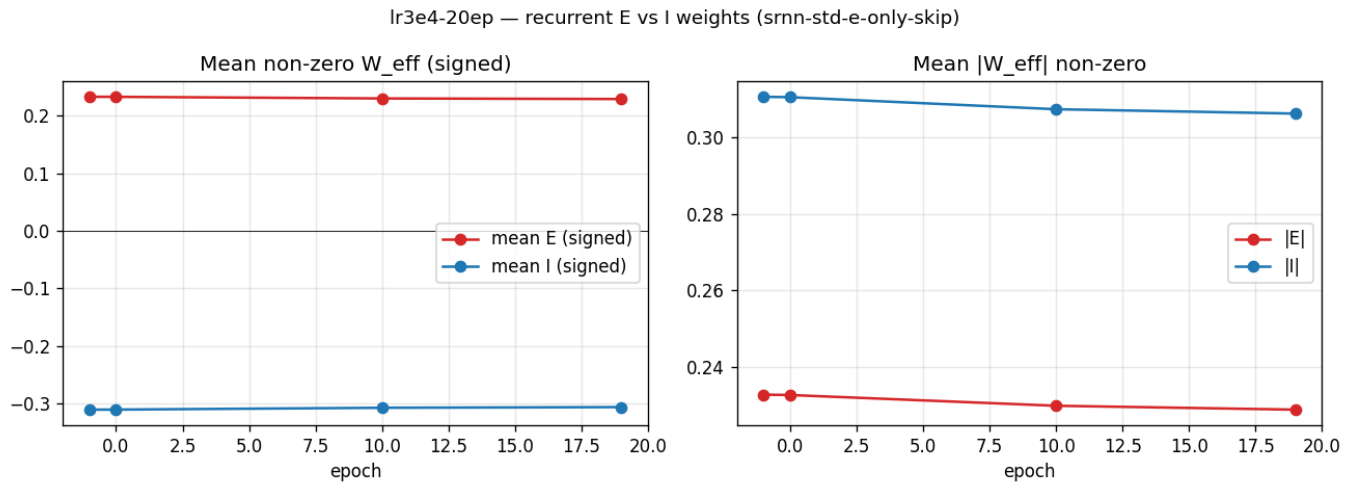


Figure 16: Tau evolution

Figure 17: W_{EI} evolution

srnn-std-e-only-skip — parameter table

lr3e4-20ep - variant srnn-std-e-only-skip (k=7)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.99784	-	-	scalar
tau_d (s)	+0.1	+0.10125	0	0	300
tau_a_E (s)	+0.69315	+0.69165	0	0	450
tau_a_I (s)	+0.69315	+0.69165	0	0	450
tau_b_rec_E (s)	+1	+1.0113	0	0	150
tau_b_rel_E (s)	+0.25	+0.24593	0	0	150
tau_b_rec_I (s)	+1	+0.99784	0	0	150
tau_b_rel_I (s)	+0.25	+0.24946	0	0	150
W_{eff} (E src, signed)	+0.23267	+0.22874	0.07641	0.07516	15048
W_{eff} (I src, signed)	-0.31062	-0.30623	0.07724	0.07618	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.22874	0.07641	0.07516	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30623	0.07724	0.07618	14977
$W_{\text{in_eff}}$ (input neurons)	+0.0012099	+0.0015692	0.09923	0.0971	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.3683	0	0	300
readout_weight	+0.00034909	+0.00053166	0.06671	0.06064	6675
readout_bias	+0	-0.0004303	0	0.005114	89
ic.ic (per-variant init)	+0.15334	+0.15985	0.4128	0.4103	1500